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THE ISOLATION OF THE CEREBRAL CIRCULATION AND THE PERFUSION OF THE BRAIN IN THE LIVING CAT

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Hitherto the main obstacle to the development of a brain perfusion method in the living animal has been the failure to isolate completely the cerebral circulation. Earlier brain perfusion experiments were either performed on the severed head (Bouckaert and Jourdan, 1936; Chute and Smyth, 1939) or in others quantitative collection of the blood was impossible owing to the incomplete blockage of the venous outlets of the brain (Schmidt, 1928). It became evident at the beginning of our experiments in 1940 that the main problem in isolating the venous outflow was to gain access to and to occlude the sinusii columnae vertebralis which may carry the entire venous return from the brain, as mentioned already in 1896 by L. Hill. (See also Harris, 1941.)

This paper presents a brain perfusion method based on the complete isolation of the cerebral circulation in the living cat. Brain metabolism experiments are described in which the animals were kept alive and without visible impairment of cerebral functions for over two hours while perfusing their brain with defibrinated heparinised ox blood.

Anatomical considerations and surgical technique. Our surgical procedure consists mainly in the isolation and occlusion of the venous outlets from the brain and is supplemented by the ligature of certain arteries. In the cat, unlike the monkey (Dumke and Schmidt, 1943; Batson, 1944) the separation of the cerebral from the extracranial circulation on the arterial side is practically impossible, as most, if not all the blood from the carotids reaches the brain via anastomoses between the rete mirabile (Ask Upmark, 1935) of the arteria maxillaris interna and the circle of Willis. The internal carotid is an insignificant vessel, or, according to Davis and Storey (1943) has no lumen at all. The anatomy of the venous outlets from the cat's brain is described here in some detail, not as an original contribution but rather as a reminder in view of some of the past errors which have arisen due to its neglect. Since the completion of our anatomical observations, Batson (1944) has described many of these anatomical details.

Each of the two sinusii columnae vertebralis is formed in the atlas by the confluence of two veins issuing from the cranium through the foramen magnum, one dorsally, the other ventrally. Thus, including the vertebral veins, at the level between the occiput and the atlas, there are three pairs of veins with free communication among themselves (within the cranium and the atlas) to ensure effective drainage of the cerebral sinuses. The "spinal venous sinuses" (sinusii columnae vertebralis) run the length of the vertebral column communicating

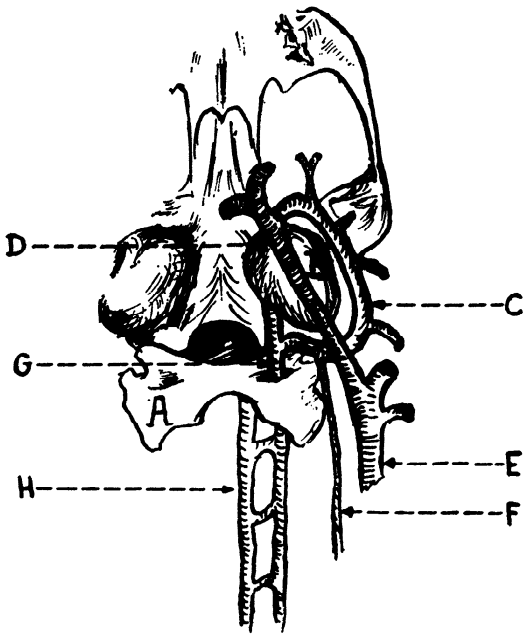


Fig. 1

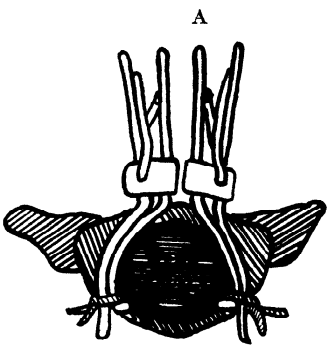


Fig. 2

Fig. 1. The branch connecting the vertebral vein with the external jugular vein and the communications between the vertebral vein and the sinus columnae vertebralis. *A*—atlas; *B*—bulla tympani; *C*—communicating branch between the vertebral and posterior facial veins; *D*—posterior facial vein; *E*—external jugular vein; *F*—internal jugular vein; *G*—vertebral vein; *H*—sinus columnae vertebralis.

Fig. 2. The position of the clips compressing the sinusii columnae vertebralis between atlas and occiput shown in cross-section. *A*—ventral side.

at *each* interspace with the vertebral veins (as shown schematically in fig. 1). After the exit of the latter from the transverse foramen they communicate in their further course with the Azygos system by way of the intercostal veins. Thus, unless the sinuses and the vertebral veins are occluded at one and the same level, blood will escape from one pair of veins to the other by means of their communications at other segments. The spinal venous sinuses also communicate with each other by means of small dorsal connecting branches.

After leaving the skull and before entering the transverse foramen in the atlas the vertebral veins give off a large branch connecting them with the external

jugular veins (fig. 1). This branch after giving off the internal jugular[1] and communicating with its homologue on the opposite side, circles the bulla tympani (receiving in its course 3–4 other small branches) and finally has its outlet into the posterior and/or anterior facial branch of the external jugular vein. We are not aware that this branch has been described elsewhere. Thus, if the external jugulars are ligated, all the extracranial venous blood may enter the vertebral veins by means of this branch, which may carry blood in either direction. In addition to the important venous exits just mentioned there are other small venous communications through the trigeminal fissures between the brain and extracranial tissues. These are excluded from the circulation by occluding the arteries which feed the tissues, from which they originate.

The following branches of the carotid arteries are ligated: a. The superior thyroid and dorsal muscular branches of the common carotids. b. The lingual, external maxillary and submaxillary branches of the external carotids. c. The terminal section of the internal maxillary artery rostral to the anastomotic branches supplying the circle of Willis. This last ligation is performed through the mouth, the artery easily being exposed in the angle formed at the posterior junction of the palatine and maxillary bones and tied caudal to the bifurcation of the aa. infraorbitalis and sphenopalatina.

The vertebral arteries are ligated before their entrance into the transverse foramen of the 6th vertebra. Though there are numerous small muscular branches arising from the vertebral arteries as they course upwards, we have never lost perfusion blood through these branches, possibly because the external jugulars, which collect a large proportion of the muscular venous return from this region, are occluded. A small branch which forms an anastomosis between the vertebral and carotid arteries is ligated together with the vertebral vein on each side. From our experience with a very large number of animals we are satisfied that the anterior spinal artery in the cat is so small that it may always be safely disregarded.

On the basis of numerous injections and perfusion experiments, we may safely conclude that the method described here isolates effectively the cerebral venous outflow so that all the venous blood from the brain may be collected, as described subsequently, with the possible admixture of but insignificant amounts from extracranial tissues.

The technique of the operation is as follows: The cat (preferably lean, short-haired and weighing about 3 kgm.) is first anesthetized with ether and then given 70 mgm. of Dial/kgm. intravenously. After the insertion of a tracheal cannula the vertebral arteries and the above mentioned branches of the carotids are ligated, the trachea and esophagus are cut and retroflexed and the anterior occipito-atlantoid ligament and vertebral veins at this level exposed. The branches connecting the vertebral and external jugular veins are tied. The

[1] In cats the internal jugular veins are usually very small. In about $\frac{1}{5}$ of our cats, one larger internal jugular was found, usually on the left side, and in these cases the vertebral vein on the same side was invariably smaller than usual.

animal is then turned over and the dorsal occipito-atlantoid ligament exposed. The capsule between occiput and atlas is cut as far as possible on both sides. The occipital bone is scraped clean for subsequent insertion of cannulae. The cat is turned over on its back again and a longitudinal incision is made through the middle of the anterior occipito-atlantoid ligament and underlying dura. Both the ventral and dorsal pairs of veins which go to form the spinal sinuses are compressed by specially constructed spring clips as shown in figure 2.

The inner jaw of the open clip is slid between dura and spinal cord through the ventral incision. The outer jaw lies between the atlas and the occiput external to the lateral part of the occipito-atlantoid ligament with its tip emerging dorsally; after closing the clip the cat is again turned over, the dura incised dorsally thus exposing the small hook at the end of its inner jaw to which the protruding end of the outer jaw is firmy tied, with strong silk thread. After administration of heparin to the animal (1000 units/kgm.) a hole is bored through the occipital bone on each side, just above and a little to the side of the occipital condyle thus tapping the transverse sinuses of the occipital bone which in the cat collect the blood from all the cerebral sinuses. Some practice is necessary for the proper placing of these holes and care must be taken not to pierce the meningeal layer of the dura. Metal cannulae are screwed into the holes. (After numerous attempts at using the torcular Herophilii for collection of the venous blood this was given up as it is not sufficiently developed in the cat.) After ascertaining that blood is flowing freely through the two cannulae just inserted, the cat is turned on its back and the cannulae are connected to an external jugular vein thus ensuring adequate venous outflow from the brain until the start of perfusion. T-cannulae are inserted into both carotids and the remaining jugular vein, and the vertebral veins are ligated between occiput and atlas. After connecting the arterial and venous cannulae to the perfusion apparatus, perfusion is started and the systemic connections of the carotids and of the jugulars clamped. By changing the position of the clamps it is easy to re-establish the animal's own circulation to the brain. The systemic blood pressure of the animal is recorded with a membrane manometer from a femoral artery and respiration by means of a pneumograph. After finishing the experiment the brain is removed down to the calamus scriptorius and weighed. At this level the sinusii columnae vertebralis were occluded.

The apparatus. The brain perfusion apparatus is shown schematically in figure 3. The following parts require special description.

The pressure regulator (fig. 4 and *H* in fig. 3) consists of two chambers: *A* and *B*, separated by a membrane about 2 mm. thick made of two sheets of rubber glued together, with a thin steel plate inserted between them at the centre. The upper chamber *A* is connected to the central stump of the carotid artery or to a manometer and syringe through which pressure can be exerted. The lower chamber *B* contains the inlet and outlet tubes (*c*) of the perfusion circuit and an overflow (*d*), the mouth of which is ground to the level of the rim of the lower chamber *B*.

Blood from the pump entering the lower chamber B at a pressure exceeding that in A, lifts the membrane and opens the overflow, equalising the pressure

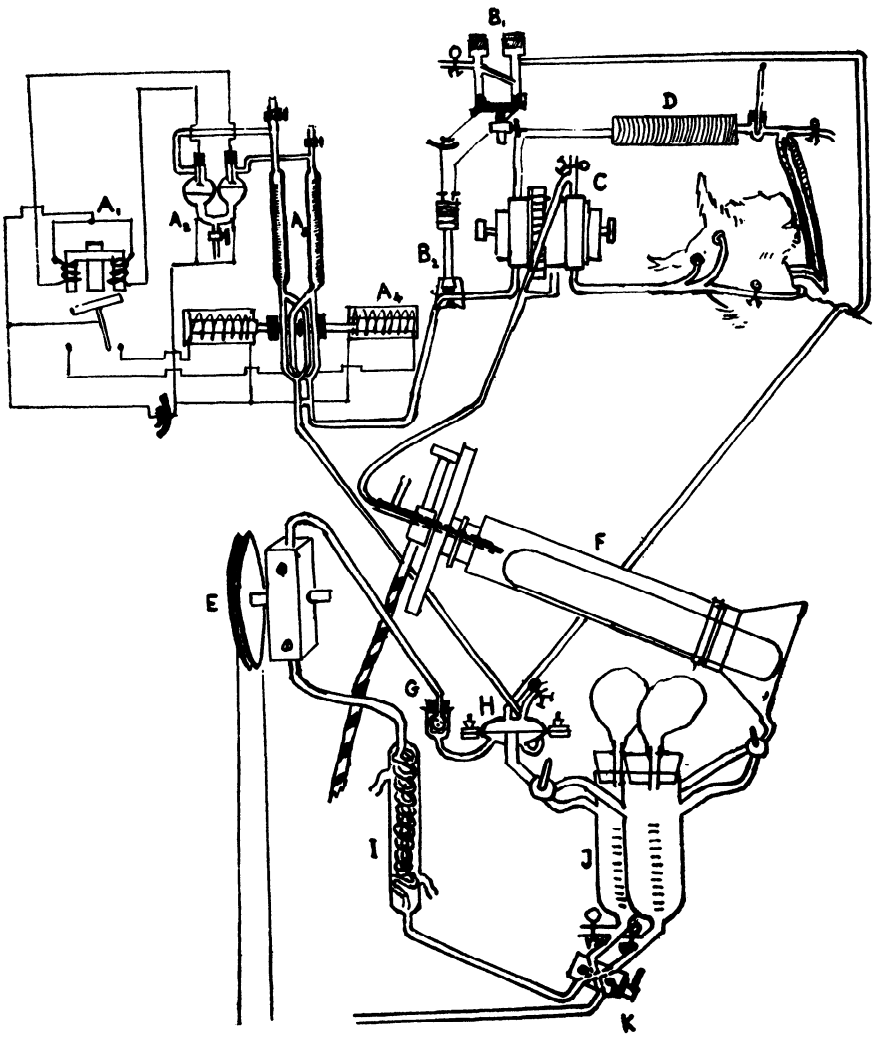


Fig. 3. Diagram of brain perfusion apparatus. A_1—Flowmeter, polarized relay. A_2—Flowmeter, electrolytic interrupter. A_3—Flowmeter, graduated blood pipettes. A_4—Flowmeter, electromagnetic valve. B_1—Pulse frequency regulator, interrupter. B_2—Pulse frequency regulator, electromagnetic valve. C—Photoelectric apparatus for measuring A-V oxygen differences. D—Heater. E—Roller pump. F—Oxygenator. G—Blood filter. H—Pressure regulator. J—Graduated blood reservoirs. K—Stopcock.

in both chambers. As the upper chamber is connected to the systemic end of the carotid artery and as the systemic blood pressure is still under central control, vasomotor impulses from the brain are able to regulate the perfusion pressure.

The inside diameter of the pressure regulator is 40 mm. and that of the overflow tube 5 mm. The diameter of the overflow tube should be small in proportion to that of the membrane in order to minimise the error due to the fact that the membrane area covering the overflow tube is under pressure only from chamber *A*. Thus the pressure in chamber *B* will exceed that in chamber *A* in proportion to the fraction: surface area of overflow tube/surface area of membrane. In our apparatus with a membrane area of 12.5 cm^2, and an overflow area of 0.2 cm^2 the pressure difference between chamber *A* and *B* is less than 2 per cent.

The flowmeter (*A* in fig. 3) is essentially the instrument described by Montgomery et al. (1934), with electrolyte control of electro-magnetic valves operated by a 10 volt current through a polarised relay. In our apparatus two graduated pipettes are used as blood chambers and the blood is separated from the electrolyte (saturated KCl solution) by air.

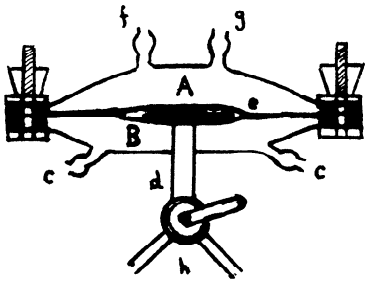


Fig. 4. Blood pressure regulator. *A*—upper chamber; *B*—lower chamber; *c*—inlet and outlet tubes for blood; *d*—overflow tube; *e*—rubber membrane with metal disc in center; *f*—tube connecting upper chamber to central stump of carotid artery; *g*—air outlet tube; *h*—stopcock.

The pulse-frequency regulator (*B* in fig. 3) is an electromagnetic valve, which clamps down on soft rubber tubes with every diastole and releases them on systole. This valve is activated through a U-tube, partly filled with mercury. A fine capillary tube connects its arms in order to equalise pressures when there is a change in the animal's blood pressure. This valve is activated by the animal's pulse or by a mechanical interrupter.

The oxygenator (fig. 5) is built on the principle of a Barcroft tonometer. Its main parts are two rotating cylinders of glass or tin-plated brass, one inside the other. The inner cylinder increases the oxygenating surface and distributes the blood in a thin uniform layer.

The instrument is held in position at a slant of about 25 degrees in a rocking joint at its upper end and by a rod turning freely in a tubular holder at the lower end. It is revolved at about 40–50 r.p.m. by a friction wheel connected to a flexible shaft. This oxygenator which is 50 cm. long and has a diameter of 6 cm. is capable of introducing 5 ccm. of oxygen per minute into 100 cc. of venous blood flowing at a rate of about 200 ccm. per minute. It may be built in different sizes according to requirements.

The photoelectric apparatus for continuous measurement of the oxygen content of the blood is a modification of the apparatus described by Kramer and Winton

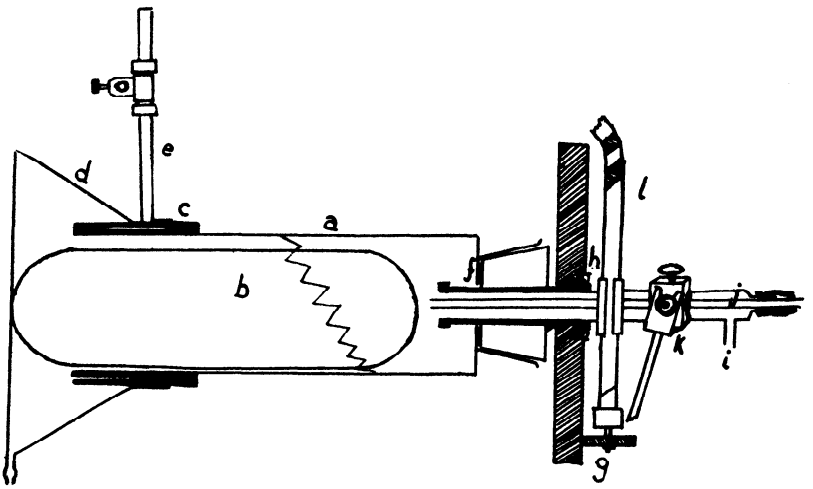


Fig. 5. Oxygenator. *a*—outer cylinder; *b*—inner cylinder; *c*—bearing; *d*—collecting funnel; *e*—swivel joint holder; *f*—rubber packing; *g*—friction wheels; *h*—ball bearing; *i*—gas inlet tube; *j*—blood inlet tube; *k*—rocking joint; *l*—flexible shaft.

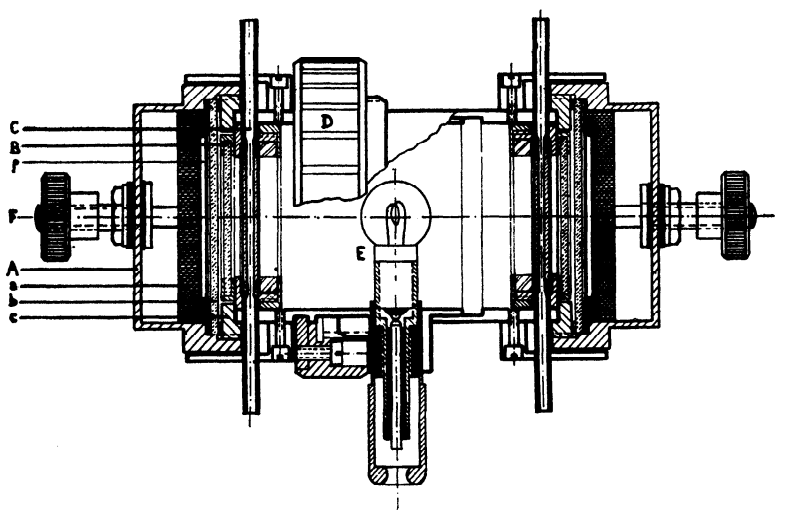


Fig. 6. Photoelectric apparatus for measuring A-V oxygen differences. *A*—metal housing containing photocell; *a*—hard rubber backplate holding photocell P and contacts; *b*—glass plate between rubber gaskets; *c*—threaded metal ring sealing photocell unit; *B*—red filter; *C*—cuvette; *D*—large adjusting screw for regulating distance of second photocell from light; *E*—6 volt, 0.5 amp. electric bulb in adjustable holder; *F*—photocell terminal.

(1939) and is constructed so that continuous and direct determination of arteriovenous oxygen differences can be made (fig. 6).

This apparatus makes possible the measurement of the respiratory metabolism of the brain even under rapidly changing conditions and dispenses with a large

number of blood gas determinations. It consists of two symmetrical halves, each containing a barrier layer photocell, a red filter (6200–6800 A°) and a new type cuvette through which the arterial and venous blood flow. The cuvettes are constructed so that the thickness of the central plate (*d* in fig. 7), which determines the thickness of the blood layer in front of the photocells, is identical in both halves. Thus the arterio-venous oxygen differences may be measured directly by connecting both photocells in compensation. Dismantling of the cuvettes, cleaning and reassembling can be carried out without changes in their thickness. The area of the slit in the central plate of the cuvette determines the area of illumination of the photocell. The flow velocity of the blood through the cuvette should be high enough to prevent sedimentation of the blood corpuscles which increases light transmission through the blood and may cause errors up to several hundred per cent. This can be prevented by a proper choice

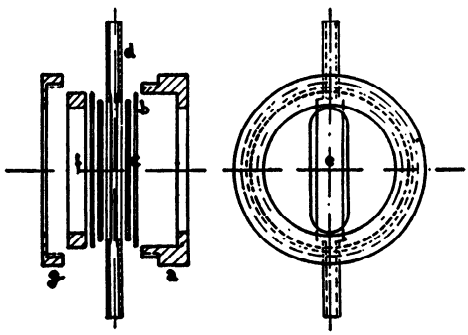


Fig. 7. Blood cuvette of photoelectric apparatus. *a*—metal housing; *b*—rubber gaskets; *c*—circular glass windows, *d*—central plate ground plane-parallel, 1.2 mm. thick; *e*—6 mm. wide slit in central plate through which blood flows; *f*—metal compression ring; *g*—threaded sealing ring.

of the dimensions of the slit in the central plate. In our apparatus the slit is 6 mm. wide and the central plate 1.2 mm. thick. Although the sedimentation rate of cat blood corpuscles is very high, measurements of arterio-venous oxygen differences with this apparatus were still reliable at a flow rate as slow as 5 ml. per minute. In order to translate galvanometer deflection into A-V-O_2 differences calibration curves were constructed using blood samples of different hemoglobin content. The measurements were made with the photocells connected in compensation and with the light, galvanometer and scale arranged so that a deflection of about 50 mm. corresponded to an A-V oxygen difference of 1 vol. per cent. It was found that by adjusting the light intensity so as to give the same galvanometer deflection (in our experiment 600 mm.) with every arterial sample, the same calibration curve could be used for any blood sample, having an oxygen capacity between 12–16 vol. per cent.

The liver perfusion circuit for removal of vasoconstrictor substances from the blood consists of a roller pump which conveys the blood from a large reservoir into the liver and back. The perfusion pressure is regulated by the height of an overflow vessel above the liver. About 2 hours before starting brain perfusion

the liver of a freshly killed cat which has been given a small amount of heparin and bled to death under ether, is placed in the apparatus with a cannula in the portal vein. The hepatic artery is tied and the hepatic veins are cut so as to allow free drainage from the liver. After washing out the liver with saline, perfusion is started through the portal vein with well filtered defibrinated ox blood to which a small amount of heparin has been added. The blood is passed through the liver at least 4 times after which it is transferred to the main perfusion apparatus and saturated with the desired gas mixture. During brain perfusion the blood does not come into contact with the liver.

The minimum amount of blood required in order to take maximum advantage of a perfusion experiment is about 600 ml. This includes about 250 ml. for analysis in an experiment of long duration, 150 ml. for one filling of the perfusion apparatus and a reserve of 200 ml. To have this amount of cat blood on hand would mean bleeding about 6 additional cats for each experiment. Consequently after a number of trials with defibrinated beef blood from the slaughter house, it became apparent that this could serve as well. The blood is usually collected the day before the experiment, filtered well through cloth and kept in a refrigerator. When kept for longer than 24 hours, Merthiolate-Na (Lilly) 1:20,000 is used as a preservative. We have never had any reactions from the use of bovine blood in the cat. Ox blood with its original high red blood corpuscle content offers excessive resistance to flow through the brain. Accordingly, the blood used for perfusion was diluted with plasma to give an O_2 capacity around 14 vol. per cent. We have seldom found a higher oxygen capacity in the blood of narcotised cats and very often it has been even as low as 8 vol. per cent.

In our early experiments the blood was perfused before and during brain perfusion through the lungs of a freshly killed cat in order to get rid of the vasoconstrictor substances which develop in the shed blood upon standing. The very temporary effect of this procedure and the rapidity with which the lungs themselves became edematous, necessitated a search for a more efficient means of removing these v.c. substances. We found that by passing the blood several times through an isolated liver, it is possible to maintain a good blood flow through the brain indefinitely. In passing the liver, the blood will take up certain substances, thus the likelihood remains, that in addition to removing v.c. substances from the blood, the liver may in other ways be a vital factor in determining the success of brain perfusion.

Analytical methods. During experiments A-V oxygen differences are recorded continuously by the photoelectric apparatus described above. Samples for glucose, lactic acid CO_2 and pH are taken from the arterial and venous sides of the perfusion system. Glucose is determined according to Somogyi (1937), lactic acid according to Friedemann and Graeser (1933), CO_2 with the Van Slyke method and pH with the glass electrode.

EXPERIMENTAL RESULTS. The metabolism of the living cat's brain was recorded continuously while perfusing it with defibrinated heparinized ox blood. In some cases it was possible to keep the animal in excellent condition for over

two hours. At the start of perfusion a large part of the narcotic is washed out of the brain by the perfusing blood, the first 50–70 ml. of which are discarded. The small amount of narcotic remaining in the brain is greatly diluted by the perfusing blood. Thus, a few minutes after switching over the blood circulation of the brain to the perfusion apparatus, the animal seems to regain consciousness, showing perception of objects approaching its eyes, very active blinking reactions to light, rapid pupillary and corneal reflexes and spontaneous movements of the eyes and facial muscles. Respiration is good though rather fast, vasomotor and respiratory reflexes are very active and sytemic blood pressure is fairly high throughout the good part of the experiments. At the same time the rate of blood flow through the brain and the oxygen consumption are comparatively high. On the other hand spinal reflexes remain slow as the cord is still under the influence of the narcotic.

Oxygen consumption of the brain. For convenience of presentation we divided the experiments into three groups according to the functional state of the brain and the behaviour of the animal during perfusion.

A. The first group comprises the best experiments in which cerebral activity was excellent and the animal seemed to be conscious as described at the beginning of this chapter.

B. The second group consists of those experiments in which spontaneous movements and other signs of near-consciousness were absent but ocular responses to touch and sometimes to light, and vasomotor and respiratory reflexes were good. As the pupils were constricted in most experiments of this group, the pupillary reflex could not be elicited.

C. To the third group belong those experiments or experimental periods in which reflex responses were very weak or absent. There was no spontaneous respiration and the systemic blood pressure was very low. The eyeballs protruded and were glassy and the pupils were extremely dilated.

In table 1 the oxygen consumption in each of these groups is shown, each experiment being divided into fifteen minute periods. The entire course of a typical experiment belonging to group A is shown in figure 8. In this experiment cerebral activity was excellent and the animal seemed to be conscious. The blood flow through the brain was high from the beginning of perfusion, well over 100 ml. per 100 grams brain per minute, with a perfusion pressure of about 120 mm.Hg. In this experiment as in most of this group, after the first 4–6 minutes of perfusion, blood flow rate through the brain increased spontaneously, perfusion pressure remaining constant, reaching rates up to 150 ml. per 100 grams per minute.[2]

The oxygen consumption of the brain was very high at the beginning of perfusion, sometimes 6 ml. of oxygen per minute. This high oxidation rate declined within 5 to 10 minutes, as shown in table 1, to a steadier level of oxidation between 4 to 5 ml. of oxygen. Oxygen consumption was maintained at a level usually over 4 ml. as long as cerebral activity remained excellent, in some cases

[2] Subsequently the figures for blood flow and for oxygen consumption are calculated for 100 grams brain per minute.

The Oxygen consumption of the perfused brain (ml. O_2 consumed per 100 grams brains per min.). Division into 3 groups according to functional state of brain

EXPERIMENT	DURATION OF PERFUSION, MINUTES													REMARKS
	0-15	15-30	30-45	45-60	60-75	75-90	90-105	105-120	120-135	135-150	150-165	165-180	180-195	
Group A														
23-5	4.65	4.65	4.50	4.80	4.65	4.42	4.35	3.78(1)	3.45	3.15(2)	3.22	3.34	3.34	(1) Added 60 mgm. % chloral hydrate. (2) Changed blood
30-5	4.75	4.23	3.60	3.30	3.08	3.22	3.15	3.00(1)	2.70	2.48	2.85*			(1) Changed blood—strong vasoconstriction
6-6	4.57	3.98	3.74	3.45(1)	3.08*	2.80								(1) Changed blood—vasoconstriction
13-6	5.45	4.20	4.12	4.42	3.90*	2.70								
8-7	5.55	5.06	5.32	4.95	4.70	3.90(1)	3.00	2.85*						(1) Added 20 mgm. % Na-amytal
15-7	4.73	4.50	4.87	4.75	4.65	4.42(1)	4.42	3.75(2)	3.00	2.85*				(1) Changed blood. (2) Added 20 mgm. % Na-amytal
Average.	4.95	4.44	4.36	4.28	4.01									
Group B														
30-1	3.68	3.75	3.38*	3.15										
4-2	4.27	3.45	3.67	3.67*	3.45									
7-2	4.35	4.10	3.00	2.93*	2.55	2.40								
17-2	3.82	3.30	3.15*	3.15										
25-3	3.45	3.52	3.00	3.00*	2.65	2.40								
17-6	3.45	3.38	3.45*	3.22(1)	2.92									(1) Added 20 mgm. % Na-pentothal
Average.	3.83	3.58	3.27	3.18	2.90									
Group C														
25-2	3.22*	2.93	2.78	2.70	2.70									
28-2	2.55	2.40	2.10*	2.03	2.10									
4-3	3.00*	3.15	2.82	3.15										
11-3	2.32*	2.32	2.22	2.22	2.10									
21-3	3.75*	3.00	2.92	2.63	2.40									
3-7	3.75*	2.81	2.64	2.66										
Average.	3.10	2.77	2.58	2.56	2.32									

* Loss of reflexes.

for periods up to 2 hours. Subsequently oxygen consumption slowly declined together with a progressive deterioration of cerebral activity. The experiments

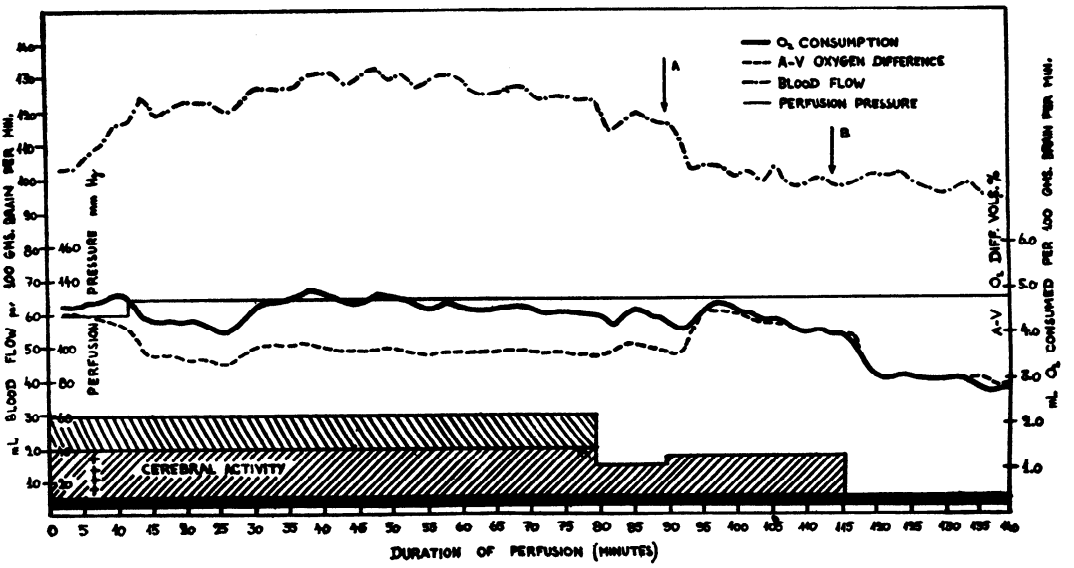


Fig. 8. Perfusion experiment with excellent cerebral activity (group A). At the bottom of the figure the uppermost crosshatched area indicates persistence of cortical activity, i.e., apparent consciousness, spontaneous movements, etc. The opposite crosshatching below it shows degree of reflex response as scored by four crosses. The thick black line above the abscissa shows the persistence of natural respiration. At *A* the perfusion blood was changed. At *B* 20 mgm. of Na-amytal were added to each 100 cc. of perfusion blood.

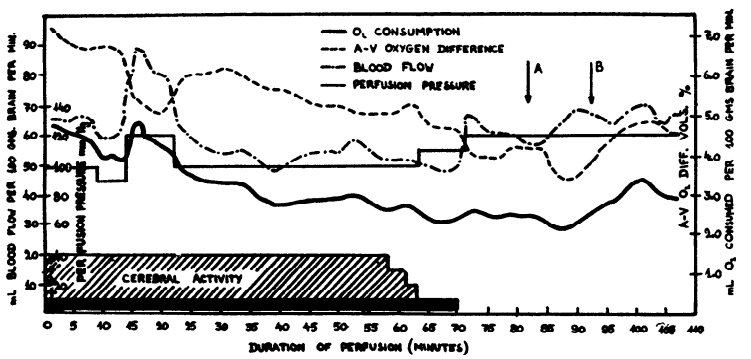


Fig. 9. Perfusion experiment with impaired cerebral activity (groups *B* and *C*), cortical activity absent. The first half of this experiment up to about 55 minutes is classified as belonging to group *B*, the latter part to group *C*. The duration of natural respiration is shown by the thick black line above the abscissa. At *A* 2 cc. of a 30 per cent NaCl solution were added to each 100 cc. of perfusion blood. At *B* 3 cc. of neutral M/5 phosphate buffer were added to each 100 cc. blood.

of this group were usually discontinued shortly after the disappearance of spontaneous activity. From numerous observations made on animals during perfusion experiments we feel justified in ascribing the observed high rate of oxygen

consumption in the experimental group A to cortical activity. In group B signs of cortical activity were absent though reflex activity was excellent and correspondingly a rate of oxygen consumption between 3.2–3.8 ml. was found.

There is a quantitative difference in their respective *responses to narcotics* between the cats in groups A and B. The addition of 20 mgm. of amytal per 100 ml. of blood abolished every sign of cortical activity spontaneous movements and ocular reflexes in an experiment belonging to group A though leaving natural respiration intact. At the same time the rate of oxygen consumption

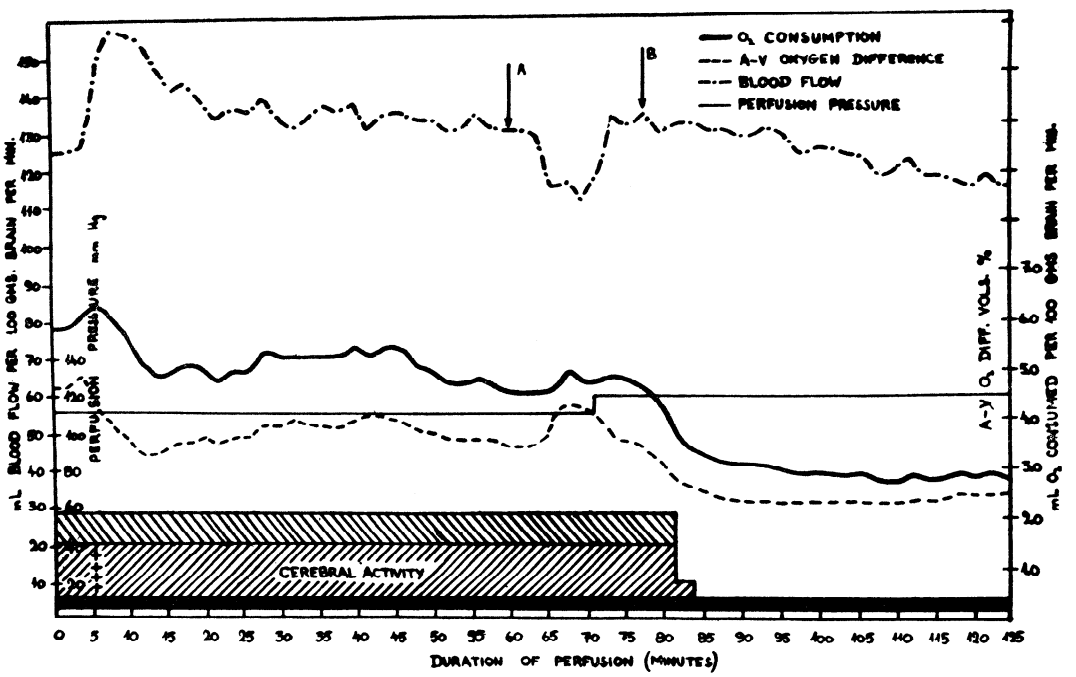


Fig. 10. The effect of narcosis on the oxygen consumption of the brain in an experiment with excellent cerebral activity (group A). Upper crosshatching shows cortical activity, thick black line natural respiration. At *A* blood was changed, at *B* 24 mgm. of Na-amytal were added to each 100 cc. of blood.

dropped from 5.1 to 2.9 ml. (43 per cent) as shown in figure 10. On the other hand the addition of 24 mgm. of amytal to the perfusion blood of a cat without cortical activity caused a reduction of oxygen consumption from 3 ml. oxygen to 2.5 ml. (17 per cent) only, as shown in figure 11.

Hyperexcitability of the cortex during strychnine and metrazol convulsions was always accompanied by a sudden very great rise of the oxygen consumption by the brain which set in together with the commencement of convulsions. The addition of barbiturates at once reduced the oxygen consumption to the previous level. Two experiments are shown in figures 11 and 12, one with strychnine, the other with metrazol. The addition of these drugs in concentration insufficient to cause convulsions or, when owing to the state of the brain convulsions could not be obtained, did not alter the rate of oxygen consumption.

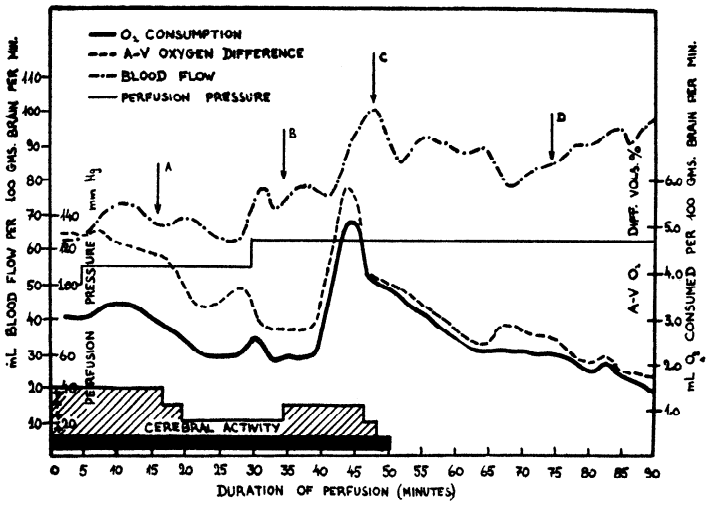


Fig. 11. Experiment, group B. At *A* 24 mgm. of Na-amytal, at *B* 2 mgm. of strychnine nitrate were added to each 100 cc. of blood. Convulsions occurred between the 38th and 48th minutes. At *C* the perfusion blood was changed. At *D* 22 mgm. of Na-amytal were added to each 100 cc. blood after all reflex activity had disappeared (group C).

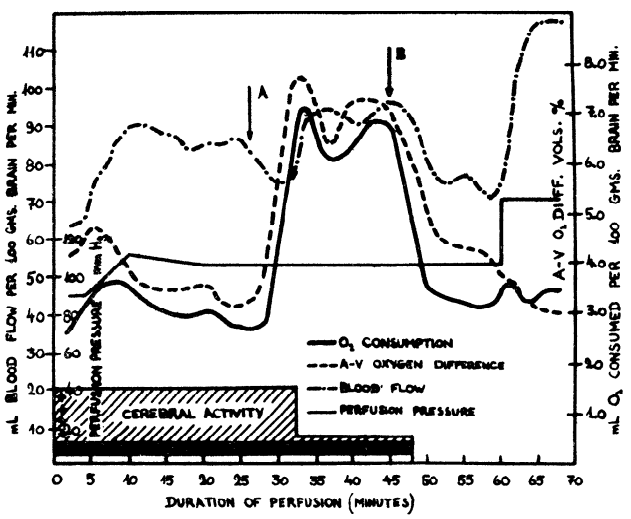


Fig. 12. Metrazol convulsions and their abolishment by pentothal (exper. group B). At *A* 60 mgm. of metrazol were added to each 100 cc. of blood. Convulsions occurred between the 28th and 48th minutes. At *B* 10 mgm. of Na-pentothal[3] were added to each 100 cc. of blood.

The rate of blood flow is an important factor in determining the oxygen consumption of the perfused brain. When the flow rate is high, between 100 to 150 ml. per minute (per 100 grams brain), increasing it affects oxygen consumption

[3] We are indebted to The Abbott Laboratories, North Chicago, Ill. for a generous supply of Pentothal Sodium.

but slightly. There is, however, a decrease in the oxygen consumption with decreasing flow rate even when the flow rate remains high. This is illustrated in figure 13. On the other hand, when the flow rate is below 80 to 90 ml. the oxygen consumption of the brain changes with every change in the rate of blood flow. In order to obtain maximum oxygen consumption, the flow rate has to be kept higher than 90 ml. per minute per hundred grams brain. Below this rate the blood flow is a factor determining the rate of oxygen consumption of the brain. Schmidt, Kety and Pennes (1945) measuring the blood flow and oxygen consumption of the monkey's brain also found that flow rate greatly influenced

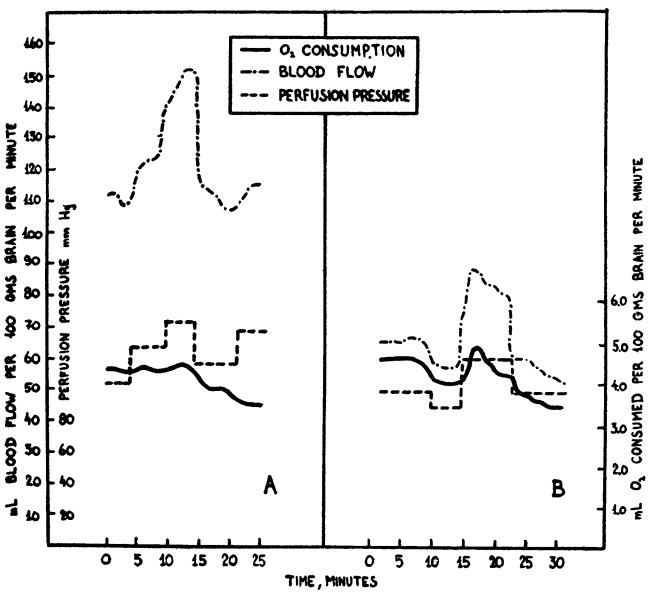


Fig. 13. Effect of changes in blood flow on oxygen consumption. A—at a high flow rate B—at a low flow rate.

its oxygen consumption. In their experiments the flow rates varied between 33 to 72 ml. per 100 grams brain per minute.

In all our best experiments after varying periods of perfusion, the oxygen consumption dropped and brain activity deteriorated. The addition of the known activators and substrates of brain carbohydrate metabolism has so far proved ineffective in restoring brain activity and oxygen consumption. A slight improvement of short duration could be obtained by raising the perfusion pressure. On the other hand the addition of fresh blood brings about a considerable increase in oxygen consumption together with improved brain activity, which improvement, unfortunately, is of short duration due to a simultaneous decrease in blood flow rate caused by the vasoconstrictor substances present in the freshly added blood. The slow deterioration of brain functions after a long period of perfusion with the same blood, and the improvement caused by the addition of fresh blood, seem to indicate that some substance which is indis-

pensable for the maintenance of normal brain functions is used up during perfusion.

The effect of CO_2 tension, of the addition of $NaHCO_3$ or of lactic acid was investigated in a large number of experiments. It appears that as far as these substances have any effect on the oxygen consumption of the brain, they exert it by altering the rate of blood flow. Thus, their effect on oxygen consumption was always marked in experiments in which the flow rate was originally low. A rise in flow rate is obtained by the addition of lactic acid, a decrease by the addition of sodium bicarbonate.

TABLE 2

The glucose consumption of the perfused brain

EXPERIMENT	DURATION	GLUCOSE		PER 100 GRAMS BRAIN PER MINUTE		% GLUCOSE ACCOUNTED FOR BY OXIDATION
		Mgm.% in blood at start	Mgm. uptake by brain	Glucose mgm.	Oxygen ml.	
	minutes					
1	33	131	41.7	4.85	3.40	95
2	25	154	34.0	4.95	2.88	78
3	33	174	72.2	8.00	3.50	58
4	16	182	17.8	3.90	3.60	123
5	28	192	61.4	7.80	3.10	53
6	18	210	21.1	5.10	3.40	89
7	22	192	28.8	5.70	2.60	61
8	42	206	102.5	8.70	4.65	71
9	86	149	149.0	6.60	3.56	72
10	55	181	85.8	5.70	4.12	96.5
11	38	217	53.2	5.10	3.60	94
12	25	181	28.0	4.08	4.12	34

A considerable rise in systemic pulse pressure was frequently noted when the H ion concentration of the blood perfusing the brain was slightly increased by raising the CO_2 tension or by the addition of lactic acid.

The diffusion of glucose into the brain. The uptake of glucose by the brain was studied in 23 perfusion experiments. Flow velocity and oxygen consumption were registered continuously, glucose, lactic acid and pH were determined in arterial and venous samples. Analyses were made in duplicate. Special care was taken not to change the rate of venous blood flow while drawing the samples.

The glucose uptake for a given period was calculated from the change in glucose content of the reservoir and from the arterio-venous differences which have been estimated at frequent intervals. The maximum possible glucose consumption was calculated from the amount of oxygen consumed and from the lactic acid produced, if any. Comparing the figures of the amount of glucose which disappeared with the amount of oxygen consumed (usually lactic acid was not produced), it was found in most experiments that various amounts of glucose were unaccounted for. These experiments are shown in table 2.

We cannot give at present any explanation for the discrepancies between glucose disappearance and oxygen consumption as the periods under investigation were in most cases not long enough to exclude the effect of possible fluctuations in the diffusion rate. On the other hand, frequent determinations of arterio-venous glucose differences have shown that such fluctuations exist and are especially marked when the glucose content of the perfusing blood is being changed. Figure 14 shows such an experiment in which the rate of the entrance of glucose fluctuates although the oxygen consumption remains unchanged at the same time. If the glucose content of the perfusing blood is higher than that of the

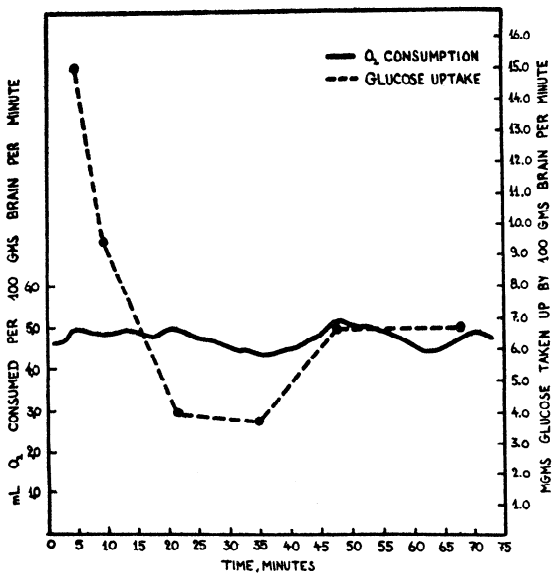


Fig. 14. Fluctuations in the glucose uptake of the brain without corresponding changes in oxygen consumption. The ordinates are so arranged that each milligram of glucose uptake (righthand ordinate) corresponds to 0.75 mL of oxygen consumed (lefthand ordinate).

cat's, the glucose uptake at the beginning of perfusion is higher than the corresponding oxygen consumption. This also happens after the addition of glucose to the perfusing blood. Conversely the venous blood may even contain more glucose than the arterial when the glucose concentration in the perfusing blood is suddenly lowered by changing the blood to one of low glucose content. In this latter case glucose is leaving the brain in order to establish a new osmotic equilibrium with the perfusing blood. From the foregoing it is obvious that the rate of glucose uptake is not identical with the rate of glucose oxidation but represents the sum of the glucose actually metabolised and that necessary to establish osmotic equilibrium.

In the hope of being able to measure the diffusion rate of glucose into the brain, several experiments were made in which a certain amount of glucose was added to the blood and, at very frequent intervals, (2 to 4 min.) arterio-venous samples

were taken for glucose determination. In these experiments a rather striking fact came to light. Only into the dead or deeply narcotised brain does glucose penetrate according to a regular and uniform pattern as shown in figure 15.

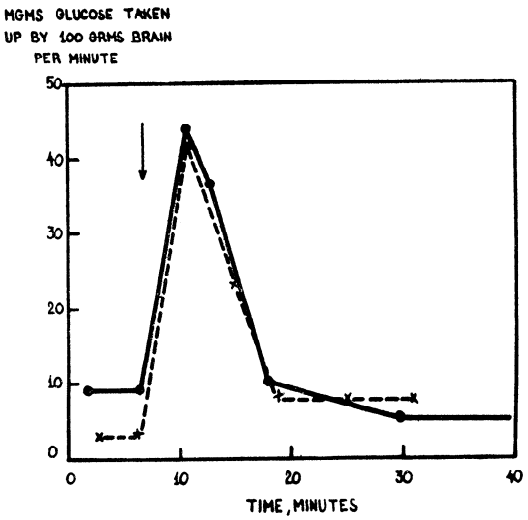


Fig. 15. The course of glucose uptake by the deeply narcotized brain (continuous line) and by the dead brain (broken line). At the arrow 100 mgm. of glucose were added to each 100 cc. of blood. The experiment with the deeply narcotized brain (continuous line) is a continuation of the experiment represented by the continuous line in figure 14, but after 20 mgm. of Na-amytal had been added to each 100 cc. of blood.

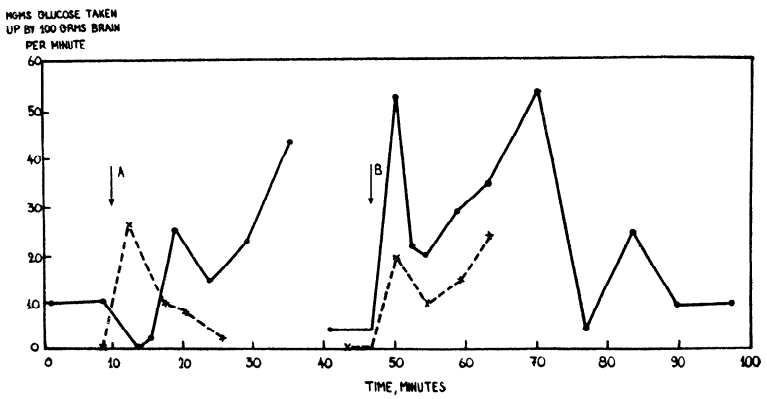


Fig. 16. The course of glucose uptake by the active brain, shown in two experiments (continuous and broken lines), after the addition of 100 mgm. of glucose to each 100 cc. of perfusion blood at *A* and at *B*.

In these experiments right after the addition of glucose there is a high rate of glucose uptake into the brain which reaches a peak during the first four to five minutes and then declines within ten to fifteen minutes to a fairly constant level. On the other hand, the diffusion of glucose into the active brain is irregular and

individual as shown in figure 16. These experiments which were made on cats with excellent cerebral activity have shown large irregular oscillations in glucose uptake while the oxygen consumption remained constant. As these oscillations did not occur in the dead or deeply narcotised cat, the assumption is made that the permeability of the brain to glucose is connected with its functional activity. This point is further illustrated by experiments on one and the same animal in which the diffusion of glucose into the brain was measured while brain activity was excellent and again after spontaneous and reflex activity have been abolished by sodium amytal. In figure 16 the glucose uptake after the first addition is very irregular, while after the addition of the narcotic it corresponds with that of the dead brain.

SUMMARY

1. The anatomical details and surgical difficulties involved in isolating the cerebral circulation are described. The venous outlets from the brain of the cat, particularly the sinusii columnae vertebralis and an important communicating branch between the vertebral and the external jugular veins are described.

2. A new method for the isolation of the cerebral circulation is described. It consists in occluding all venous outlets from the brain, including the sinusii columnae vertebralis with special clips, the ligation of certain arteries, and in tapping two occipital sinuses for the collection of the cerebral venous blood in toto.

3. A very effective method for the removal of vasoconstrictor substances from the blood by perfusing it through an isolated liver is described.

4. A perfusion apparatus, especially adapted for brain perfusion, is described in detail. Some of its features are:

a. A pressure regulator which is under central vasomotor control.

b. A pulse-frequency regulator also under central vasomotor control.

c. An efficient new-type blood oxygenator.

d. A cuvette for a photoelectric apparatus for direct continuous measurements of arterio-venous oxygen differences in flowing blood.

5. Experiments are described in which cats were kept alive with apparently good cerebral functions while perfusing the brain with defibrinated heparinized ox blood. In the best experiments cerebral functions were maintained in excellent condition for perfusion periods of two hours and more as manifested by near-consciousness, spontaneous movements, natural respiration and very active ocular, respiratory and vasomotor reflexes. During these experiments continuous records of cerebral blood flow and of A-V oxygen differences were made. Arterial and venous blood samples were withdrawn at suitable intervals for analysis of glucose, lactic acid, CO_2 and pH.

6. The oxygen consumption of the brain was found to be related to the degree of cerebral activity. Hence the experiments were divided into three groups according to the functional state of the brain.

In group A cerebral activity was excellent (spontaneous movements, etc.) and O_2 consumption was at its highest (4–5 ml. per 100 grams per min.). In

group B with good reflex activity but without spontaneous movements, etc. the O_2 consumption lay between 3.2–3.8 ml. per 100 grams brain per minute. In group C with very weak or no reflex activity, O_2 consumption was the lowest (2–3 ml. per 100 grams per min.). The high rate of O_2 consumption in group A is attributed to maintained cortical activity.

7. The effect of adding various barbiturates to the perfusion blood was studied. All of the substances used (amytal, luminal, dial and pentothal) depressed cerebral oxygen consumption and brain activity. When brain activity was excellent and oxygen consumption high (group A), the addition of a small amount of narcotic to the brain depressed oxygen consumption greatly and abolished all signs of cortical and reflex activity. When cerebral activity was only fair and oxygen consumption lower, adding the same dose of narcotic had a relatively much weaker depressant effect on oxygen consumption.

Amytal and luminal were without effect on cerebral blood flow whereas dial and pentothal had a vasoconstrictor effect. Administration of amytal to the perfusion blood was instrumental in raising the systemic blood pressure.

8. During metrazol and strychnine convulsions very high rates of cerebral oxygen consumption were observed and blood flow rate also rose. Convulsions could be abolished by the addition of barbiturates to the perfusion blood. When the convulsant drug was given in subconvulsant doses or to moribund animals, the oxygen consumption did not change.

9. The relation of cerebral blood flow to O_2 consumption was analysed. When the blood flow was above a level of about 90 ml per 100 grams brain per minute, changes in the rate of flow had but little effect on O_2 consumption. Below this level the O_2 consumption varied with the rate of blood flow so that at low flow rates the rate of blood flow becomes an important factor in determining the rate of C_2 consumption.

10. Contrary to expectations, none but extreme changes in the CO_2 tension or other changes in the pH of the blood (addition of $NaHCO_3$ or lactic acid) had any effect on cerebral blood flow. When there was an effect, increased pH lowered blood flow and increased acidity increased flow. Very often an increase in the H ion concentration of the perfusion blood caused a rise in the systemic pulse pressure of the cat and sometimes in the mean blood pressure as well.

11. The glucose uptake by the brain showed considerable variations when followed over fairly long periods of time, though the oxygen consumption remained constant. Frequently the amount of glucose disappearing was in excess of that which could be accounted for by oxygen consumption and lactic acid production.

Frequent determinations of A-V glucose differences showed fluctuations in the rate of glucose diffusion into the brain which were especially marked upon changing the concentration of glucose in the perfusing blood.

The glucose uptake of the brain represents the sum of the glucose actually metabolized and of that necessary to establish equilibrium between blood and brain.

The glucose uptake by the deeply narcotized or the dead brain, after the addition of glucose, followed a regular reproducible pattern. On the other hand diffusion into the active brain was very irregular and individual. It is suggested that the permeability of the brain to glucose is associated with its functional activity.

It is a pleasure to acknowledge the untiring help of Mr. Z. Tuttnauer, head of our mechanical workshops. To Mr. A. Sochazewer we are indebted for helpful suggestions in the construction of the apparatus. Finally we wish to thank Dr. S. Meiboom of The Department of Physics for his constant advice in connection with the construction of the photoelectric apparatus. The photoelectric apparatus and the clips for compression of the spinal venous sinuses were made by Mr. H. Kahn and the glass parts of the perfusion apparatus by Mr. Rosenbaum.

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